

# Woongjib Choi

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## RESEARCH INTERESTS

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Digital Signal Processing, Audio and Speech Signal Processing, Audio Compression, Generative Models

## EDUCATION

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### Yonsei University

Mar. 2024 – Feb. 2026

*M.S. in Electrical and Electronic Engineering (Advisor: Prof. Hong-Goo Kang)*

Seoul, South Korea

- GPA: 4.08 / 4.30
- Thesis: *A Multi-Rate Audio Super-Resolution Framework Using Flow Matching Without Neural Vocoders*

### Yonsei University

Mar. 2018 – Feb. 2024

*B.S. in Electrical and Electronic Engineering*

Seoul, South Korea

- GPA: 3.83 / 4.30 · Includes two-year mandatory military service (Apr. 2020 – Oct. 2021)

## RESEARCH EXPERIENCE

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### Postgraduate Researcher

Mar. 2026 – Present

*DSPAI Lab, Yonsei University (Advisor: Prof. Hong-Goo Kang)*

Seoul, South Korea

- Conducting research on **neural audio codecs**, **tokenization**, and **generative models** for audio.
- In collaboration with *Prof. Yongshik Lee* on vehicle state classification from unintentional electromagnetic emissions, including STFT-based signal analysis and DNN-based classification.

### Graduate Researcher

Mar. 2024 – Feb. 2026

*DSPAI Lab, Yonsei University (Advisor: Prof. Hong-Goo Kang)*

Seoul, South Korea

- Developed **bandwidth extension** methods via **flow matching** and **self-supervised representations**.
- **ETRI-Yonsei Joint Research Project**: Led development of DNN-based post-processing for legacy audio codecs. Served as project leader in the final 6 months, resulting in 1 int'l conference paper, 1 domestic paper, and 1 patent.

### Undergraduate Research Intern

Jun. 2023 – Jan. 2024

*SRIP Lab, Yonsei University (Supervisor: Prof. Moon Gi Kang)*

Seoul, South Korea

- Implemented classical and deep learning-based image processing algorithms including white balancing, denoising, demosaicing, and super-resolution.
- Undergraduate thesis: *A Technical Overview of Deep Learning-Based Single-Image Super-Resolution Methods*

## PUBLICATIONS

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### Peer-Reviewed Publications

- [1] **W. Choi**, S. Lee, H. Lim, and H.-G. Kang, “UniverSR: Unified and Versatile Audio Super-Resolution via Vocoder-Free Flow Matching,” in *ICASSP*, 2026. [link]
- [2] **W. Choi**, B. H. Kim, H. Lim, I. Jang, and H.-G. Kang, “Neural Spectral Band Generation for Audio Coding,” in *INTERSPEECH*, 2025. [link]

### Preprints & Submissions

- [3] S. Lee, **W. Choi**, J. Kim, and H.-G. Kang, “SAGE-LD: Towards Scalable and Generalizable End-to-End Language Diarization via Simulating Code-Switching Scenarios,” *Submitted to INTERSPEECH*, 2026. [arXiv]
- [4] S. Lee, E. Ahn, **W. Choi**, and H.-G. Kang, “UR-BERT: Scaling Text Encoders for Massively Multilingual TTS Through Universal Romanization and Speech Token Prediction,” *Submitted to INTERSPEECH*, 2026.

Domestic Publications

- [5] J. Shin, **W. Choi**, B. H. Kim, I. Jang, and H.-G. Kang, “DNN-based Speech Codec Enhancement Utilizing Conditional Flow Matching,” in *Proc. KIBME Conf.*, 2025.
- [6] **W. Choi**, B. H. Kim, and H.-G. Kang, “Non-Blind Speech Bandwidth Extension Using Pre-Trained Self-Supervised Features,” in *Proc. IEIE Conf.*, 2024.

PATENTS

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|----------------------------------------------------------------------------------------------------------------------------|--------------------|
| Encoder and Decoder for Audio Coding Using Side Information Generated from Original Signal and Method of Operation Thereof | KR 10-2025-0096305 |
|----------------------------------------------------------------------------------------------------------------------------|--------------------|

TEACHING EXPERIENCE

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Teaching Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                               | Mar. 2024 – Jun. 2025 |
| <i>Department of Electrical and Electronic Engineering, Yonsei University</i>                                                                                                                                                                                                                                                                                                                                                           | Seoul, South Korea    |
| <ul style="list-style-type: none"><li>• <b>EEE2111: Analog Electronics Lab</b> (Spring 2025)</li><li>• <b>EEE2060: Signals and Systems</b> (Fall 2024)</li><li>• <b>EEE4610: Capstone Design</b> (Fall 2024)<ul style="list-style-type: none"><li>* Advised project on unified speech enhancement bridging personalized and unconditional models.</li></ul></li><li>• <b>EEE4420: Digital Signal Processing</b> (Spring 2024)</li></ul> |                       |

AWARDS & HONORS

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|------------------------------------------------------------|--------------------|
| <b>Dean’s List</b> , Yonsei University (top 3% of college) | Spring 2023        |
| <b>Academic Excellence Scholarship</b> , Yonsei University | Spring & Fall 2023 |

SKILLS

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|--------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Programming:</b> Python, MATLAB, C/C++, Verilog           | • <b>Frameworks:</b> PyTorch, torchaudio, NumPy, SciPy, Librosa |
| <b>Tools:</b> Linux Shell, Git, Arduino, Xilinx Vitis/Vivado | • <b>Languages:</b> Korean (native), English (TOEFL 104)        |

EXTRACURRICULAR ACTIVITIES

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|                                                                                             |                       |
|---------------------------------------------------------------------------------------------|-----------------------|
| <b>Editor-in-Chief</b> , Humans of Yonsei (Student Storytelling and Interview Organization) | Mar. 2022 – Jun. 2023 |
| <b>Sergeant</b> , Republic of Korea Army (Repair Parts and Tool Supply Specialist)          | Apr. 2020 – Oct. 2021 |
| <b>Freshman Class Representative</b> , Dept. of EEE, Yonsei University                      | Mar. 2018 – Sep. 2018 |